



RSET

RAJAGIRI SCHOOL OF
ENGINEERING & TECHNOLOGY
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Environmental Science & Sustainable Engineering

102908/CO900G

Individual Technical Report

Environmental Problem - Inefficient Campus Resource Management

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ABSTRACT

Educational campuses operate very much like dynamic micro-cities, consuming vast amounts of electrical and structural resources every single day. Sadly, a massive portion of this energy is completely wasted due to "phantom loads," unoptimized building management, and a general lack of awareness. This report tackles this critical environmental problem head-on by proposing the "Eco Node Campus Portal." This is a smart, decentralized edge-computing system that uses IoT sensors to track energy use in real-time.

Instead of just displaying numbers on a screen, this portal actively intervenes. It connects directly to the university's academic timetable to automatically shut off power and HVAC systems in empty rooms. Furthermore, it introduces a highly engaging, gamified token system to reward students for actively saving energy in their dorms and labs. Ultimately, the Eco Node system provides a practical, automated, and economically incentivized way to cut our campus carbon footprint by 15-20%, saving substantial funds while fostering a greener community.

INTRODUCTION

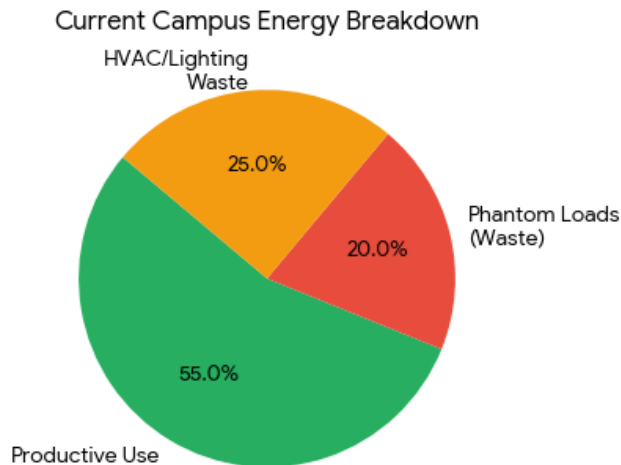
Modern engineering and academic campuses are incredible hubs of innovation, but they are also huge energy hogs. Between high-performance computer labs running complex simulations, massive lecture halls requiring constant air conditioning, and bright lighting arrays keeping pathways safe at night, we consume as much power as a medium-sized industrial factory.

The real issue we face isn't just the sheer volume of energy used; it's the fact that we don't actually see the waste happening in real-time. Getting

a massive utility bill at the end of the month tells administration how much money they owe, but it does absolutely nothing to tell us where we can improve or which specific buildings are the worst offenders. The Eco Node Campus Portal changes this dynamic completely. By utilizing modern engineering principles, it gives us real-time data and actually acts on it—turning off systems automatically when they aren't needed and transforming energy conservation into a campus-wide team effort.

PROBLEM STATEMENT

We are currently wasting energy in three primary, deeply entrenched ways. The pie chart below clearly illustrates the extent of this daily waste across the campus infrastructure.



EMPTY ROOMS, FULL POWER

One of the most glaring inefficiencies is spatial misalignment. Classes frequently get canceled, relocated, or end 20 minutes early, but the heavy lighting and central AC keep running at full capacity. Our physical buildings simply aren't talking to our digital academic schedules, leading to massive airspace conditioning waste.

BLIND SPOTS IN DATA

Right now, we only know our total aggregate campus energy cost. We don't know if a specific computer lab is drawing too much power because of a failing, inefficient AC compressor, or if heavy machinery was accidentally left overnight. This lack of granular, room-by-room data means we are operating completely in the dark.

NOBODY FEELS RESPONSIBLE

The psychological disconnect is perhaps the hardest problem to solve. Because students and academic staff don't pay the electricity bills directly out of their pockets, there's no real motivation or urgency to turn off the lights. People are completely disconnected from the financial and environmental cost of their daily waste.

SUSTAINABILITY ANALYSIS

To ensure this project is globally relevant and robust, the Eco Node system has been carefully designed to align with international sustainability standards.

SUPPORTING THE UN SDGS

- SDG 7 (Affordable and Clean Energy): Target 7.3 aims to double the global rate of improvement in energy efficiency. By cutting out wasted power through automation, we dramatically improve our local efficiency footprint.
- SDG 9 (Industry, Innovation, and Infrastructure): We aren't just changing policies; we are physically upgrading our traditional campus infrastructure with smart, edge-computing technology to make our buildings intelligent and resilient.
- SDG 12 (Responsible Consumption and Production): The gamified dashboard specifically attacks human behavior, encouraging thousands of students and faculty members to consume resources mindfully and actively participate in the solution.

ENVIRONMENTAL STANDARDS (ISO 14001)

Organizations striving for excellence look toward ISO 14001 for Environmental Management Systems. A core requirement of this standard is continuous performance evaluation. The Eco Node system acts as the digital backbone for this compliance by continuously

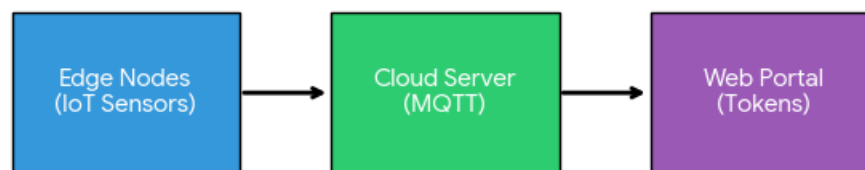
monitoring our environmental performance without requiring exhaustive manual audits.

Furthermore, by looking at the Life Cycle Assessment (LCA) of our campus, we realize that operational energy use is our biggest impact factor. By reducing how often our expensive HVAC and electrical equipment run, we actively extend their operational lifespan. This means we replace them less often, keeping thousands of pounds of heavy electronic waste out of local landfills.

PROPOSED SOLUTION: THE ECONODE SYSTEM

Instead of just building another passive dashboard that people will inevitably ignore after the first week, we are building an active, decentralized system. The workflow is entirely automated, moving seamlessly from the physical hardware up to the student portal.

EcoNode System Architecture



SMART HARDWARE AT THE EDGE

We will install low-cost, highly efficient microcontrollers (like ESP32 chips) directly at the distribution boards of major buildings. Instead of flooding the campus Wi-Fi with raw, messy data, these nodes process

the electrical noise locally—only sending essential updates to the cloud when usage spikes or drops.

LAB & SCHEDULE INTEGRATION

A custom background program will be installed on campus lab machines, automatically putting idle computers into a deep sleep mode after 15 minutes of inactivity. Additionally, the backend server will read the university timetable via API, ensuring lecture halls are only pre-cooled 10 minutes before a scheduled class starts.

THE "ECO-TOKEN" ECONOMY

Here is the most engaging part of the solution: Dorms and academic departments are given a monthly baseline energy allowance. If they consume less than their baseline, the saved kilowatt hours are converted into digital "Eco-Tokens." Students can then trade these tokens on the web portal for tangible rewards, like extra library printing credits, priority seminar hall bookings, or even discounts at the campus cafeteria. It makes saving energy highly rewarding.

TECHNICAL ARCHITECTURE

To handle the massive influx of data reliably, the technical stack must be carefully chosen.

- **Networking Protocols:** We'll use MQTT (Message Queuing Telemetry Transport). It is an incredibly lightweight protocol designed specifically for IoT devices, ensuring that our sensor nodes don't bog down the university internet bandwidth.
- **Database Schema:** Relational databases struggle with time-based metrics. Therefore, an InfluxDB time-series database will be deployed. It handles massive amounts of real-time data smoothly, allowing for instant generation of graphs on the frontend.

- Frontend UI: A fast, highly responsive, and mobile-friendly web application built in React.js will serve as the primary interface for both the students checking their token balances and the facility admins monitoring the heatmaps.

FEASIBILITY ANALYSIS

- Technical Feasibility: This project is highly viable. The hardware sensors are cheap and easily available over-the-counter. The coding languages (C for the hardware, JavaScript for the web) are well-documented industry standards.
- Economic Feasibility: The system essentially pays for itself. By actively cutting our "phantom" energy bills by an estimated 15-20%, the initial setup costs are quickly recovered. The return on investment (ROI) is conservatively expected within 14 to 18 months of full deployment.
- Operational Feasibility: Because the system relies heavily on automation, it drastically reduces the workload. Facility managers won't be bogged down with manual audits; they simply react to automated alerts

EXPECTED ENVIRONMENTAL BENEFITS

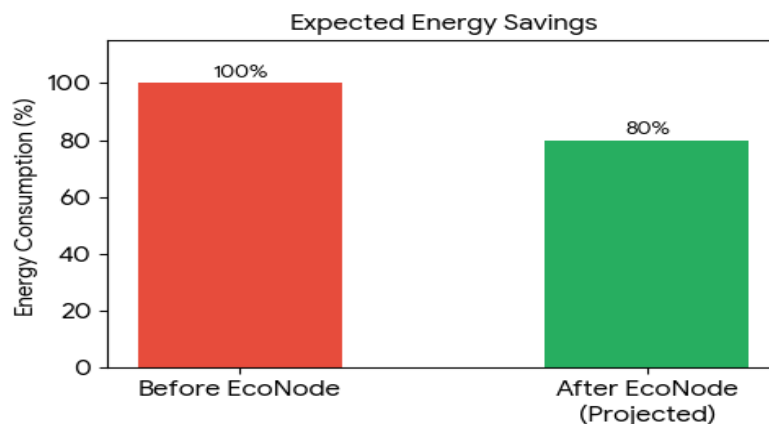
The physical implementation of the Eco Node Portal will yield measurable and immediate environmental dividends for both the institution and the surrounding community.

1. A Significantly Smaller Carbon Footprint: By actively reducing our overall grid electricity consumption, the campus will directly prevent hundreds of metric tons of CO₂ from being emitted by regional power plants every single year.

2. Enhancing Grid Stability: University campuses often trigger severe strain on municipal power grids during morning start-up phases. By staggering when heavy chillers and lab equipment turn on, we flatten the demand curve and prevent local brownouts.

3. Drastic Reduction in E-Waste: By ensuring that critical electrical systems are not overworked, we prevent complete mechanical catastrophes. Equipment lasts years longer when it isn't running 24/7, keeping heavy industrial waste out of the ecosystem.

As shown in the chart below, eliminating these phantom loads and fully optimizing the academic schedules is projected to yield massive savings compared to our current baseline:



JUSTIFICATION OF THE PROPOSED SOLUTION

Many campuses attempt to solve their energy problems by simply executing a campus-wide LED lighting retrofit. While upgrading bulbs is helpful, it absolutely does not solve the fundamental human problem of a student leaving that efficient bulb turned on in an empty room for 12 straight hours.

The Eco Node system fixes both the technological side and the human side simultaneously. By providing empirical, real-time data, it removes all guesswork from facility management. Furthermore, the tokenized economy gamifies the entire process, completely eliminating the 'stakeholder disconnect' where students feel no responsibility for the power they consume. It's scalable, it's cost-effective, and it actively builds a genuine culture of sustainability on campus that will stay with students long after they graduate.

IMPLEMENTATION ROADMAP

Our project will be rolled out in a careful, phased approach to ensure it integrates smoothly without disrupting ongoing campus operations:

PHASE	DETAILS
Phase 1 (Weeks 1-4)	Pilot Deployment: We will install the custom edge nodes in a single, high-draw computer science laboratory to test the bare-metal C data acquisition and the workstation hibernation daemons.
Phase 2 (Weeks 5-10)	Backend & Portal Dev: Development of the Time-Series Database, writing the Eco-Token ledger algorithm, and deploying the initial frontend user interface for testing.
Phase 3 (Weeks 11-16)	Campus-Wide Scaling: Roll out sensors across main distribution boards in all major department blocks and student hostels. Connect the system fully to the academic timetable of API.

Phase 4 (Week 17+)	Economy Launch: Official launch of the Eco-Token micro-economy to the student body, accompanied by awareness campaigns and redemption partnerships with campus vendors.
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CONCLUSION

Inefficient campus resource management is a deeply entrenched problem, primarily caused by a severe lack of real-time visibility and a massive disconnect between the administration paying the bills and the students consuming the resources. However, this is a completely solvable problem.

By combining intelligent, edge-computing IoT hardware with a highly engaging, gamified token economy, the Eco Node Campus Portal turns passive, energy-wasting buildings into an active, sustainable ecosystem. It slashes immediate operational costs, drastically lowers carbon emissions, and most importantly, it gets the student body actively excited about green energy. This system sets out a new modern benchmark for how educational institutions can utilize advanced software engineering to protect the environment while saving money.

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APPENDIX: WEBSITE MOCKUPS

The following section outlines the intended UI/UX design specifications for the web application, detailing the distinct views required for both the administrators and the student body.

MOCKUP A: THE FACILITY ADMINISTRATOR DASHBOARD

Target Audience: Campus Facilities Management & Maintenance Staff

This primary screen shows live campus power usage. It features a dark-themed navigation sidebar (Campus Map, Live Load, Alerts). The top header highlights three massive KPI cards: Total Campus kW (Live), Estimated Hourly Cost, and Active Maintenance Anomalies. The center of the page holds a dynamic, easy-to-read heat map format showing which building floors are glowing red due to excessive energy intensity.

MOCKUP B: THE SCHEDULE SYNC MODULE

Target Audience: System Administrators & Academic Planners

This page utilizes a split-screen view. The left pane pulls data directly from the university API, displaying the daily class timetable (e.g., 'CS101 - Room 402 - 10:00 AM'). The right pane shows the corresponding hardware toggle switches linked directly to room HVAC controls. A clear color-coded system shows exactly when rooms automatically switch from 'Active' (Green) to 'Eco-Mode pre-cooling' (Yellow) to 'Hibernation' (Red).

MOCKUP C: THE STUDENT LEADERBOARD & WALLET

Target Audience: General Student Body & Hostel Residents

A highly engaging, mobile-friendly app interface. The top widget serves as a 'Personal Wallet Card' showing the user's current Eco-Token balance with a bright 'Redeem Tokens' button. Below this is the Campus Leaderboard, ranking dorm blocks by how much energy they've saved this month, complete with gold, silver, and bronze digital badges.

Finally, an impact translation text box converts the abstract data into fun metrics: "Awesome job! Hostel Block B saved 150 kWh this week—That's equivalent to planting 4 trees!"